

VIP IP CROWD FALL 2025

Tiffany Ni

TABLE OF CONTENTS:

08/25/2025 - Brainstorming AI/Crowdsourcing Tools

08/26/2025 - Introducing Problem (Military Spouse Employment)

09/02/2025 - More Tools on Know Yourself

09/09/2025 - Class Brainstorming

09/23/2025 - Group Prototype and Discovery

09/30/2025 - PACE Tutorial

10/14/2025 - Inputs/Outputs/Features

10/21/2025 - Ideas on AI HW hint Generation

11/01/2025 - Alternative UI in PACE

11/10/2025 - Switching to FastAPI locally

11/14/2025 - Feature Integration

12/01/2025 - More Feature Integration

8/25/2025

HW for this week: Look for AI or crowdsourcing tools that could speak to the problem of assisting job seekers in the journey of articulating their skills into a resume format and finding appropriate job opportunities. A couple of bullets as to why/how these tools are interesting from an AI or crowdsourcing perspective and how they help on this journey.

1. Lightcast - Skill My Resume

- makes it possible to create resumes that show employers the full set of skills a candidate has
- beneficial for articulating skills because it helps articulate fuzzy talents into resume language that recruiters understand.

How it works:

- choose from skills and tasks related to your work experience
- optimize your resume and tailor content for specific job openings
- builds on frameworks like O*NET which is a crowdsourced database of job skills

2. ResumeFlow.ai

- an LLM-based pipeline for personalized resume generation
- helps you tailor your resume to specific job posts
- saves your time, and great proof-of-concept for AI driven tailoring

8/26/2025

Watching Presentation on Military Spouse Employment Help:

Discovery:

1. Recognize their potential.
2. Articulating their values
3. Employers may hold unconscious misconceptions of military spouses
4. Relocation (family reasons)
5. Childcare, transportation issues

Homework:

1. Find a research article on technology/ideas that address military spouse job searching
2. Dive into one of our classmate's tool and write about how it addresses the mi

Article found: <https://arxiv.org/abs/2402.06859>

LiRank is a framework that LinkedIn build to power the backend ranking of search results, job feeds, etc

- the recruiters and job platforms would integrate LiRank into their system
- this gives military spouses a more bias-reduced job searching experience
- How?
- Uses Residual DCN + Transformers (captures nuanced resume details like gaps, volunteering that is often overlooked by standard models)

9/2/2025

HW:

Any tools that help with any of the earlier stages (know yourself, planning on moving, children care?)

- how does it use AI/crowdsourcing

at least 2 ideas on how to use AI/crowdsourcing in this know yourself space

- try to have 1 idea up by friday

Week 3: Here are two tools that I found that might be helpful to the earlier stages of the Career Development Model

Career Explorer by Sokanu:

A career assessment platform powered by machine learning, psychometrics, and real-time labor market data. It helps you uncover your personality traits, interests, strengths and matches them to careers and degrees.

It builds compatibility scores by comparing your profile (personality, goals, background) with a massive, constantly updated database of career profiles.

How it uses AI/Crowdsourcing:

Its ML models are trained on over 500 million answered questions, so it learns patterns and knows what career fits look like in real life

There is a community layer where you can share insights, ask questions, and get feedback from fellow career explorers

LinkedIn Next Role Explorer:

It's an AI-driven feature within LinkedIn Learning and Premium that maps out career moves based on your current title, goals, and what looks solid in the job market.

How it uses AI/Crowdsourcing:

9/9/2025

In-class activity: Brainstorming ideas + tools on the board.

Group Ideas: ^{online}

- A Consulting Service that supports military spouses, prepares them for the workforce, guide them to learn new skills.

My add-on: The consultant could forward the spouse to a skills expert, or mentor contacts (connected to another group).

CareerBridge Session

00:12:45 [End Call](#)

Consultant Video / Chat Window

Session Notes

- Resume Review Needed
- Confidence Coaching

Quick Actions

[Forward to Skills Expert](#)

[Forward to Mentor Contact](#)

Choose Your Next Step

 **Skills Expert**

Learn or refresh professional skills to re-enter the workforce.

[Connect →](#)

 **Mentor Contact**

Match with a mentor who understands your background and can guide you.

[Meet Mentor →](#)

Consulting UI

9/23/2025

My role in my group's prototype:

Engagement & Testing Lead – designs prototype “homework activities” and helps plan how we’ll test user engagement and collect feedback. (Tiffany)

Our prototype emphasizes personal identity and confidence building before careers,

Connecting with Other 2 Prototypes

- **Crowdsourcing Tool Prototype:** Their tool allows users to **create profiles and connect with others who share similar goals**. We could integrate with it by letting spouses crowdsource advice, job leads, or peer support while still having access to a dedicated mentor. Our app could feed into their profiles so mentors' notes and resources flow into that community space.

Mentor-Centric Prototype: Their design also centers on career mentorship. We could align by differentiating roles. Theirs might focus on career pathways directly. Linking the two could create a holistic journey: start with rediscovering self, then transition into targeted career mentorship.

HW for next week:

- create visuals. create at least 1 persona. button pushed? AI?

Prototype ideas:

Homework System

I'm thinking about a micro-assignment system ("Homework Hub") with 5–15 minute tasks that translate military spouses' experience into career assets.

Each activity has: prompt → AI assist → artifact → mentor review → micro-feedback

Stages: Confidence, Identity, Career Exploration, Job-Search Readiness

Engagement test plan

Goal: See if the "homework activities" actually help military spouses feel supported and engaged between mentor sessions.

1. What we want to test

- Do spouses **start** the activities? (Are they easy enough to begin?)
- Do spouses **finish** the activities? (Are they the right size and not overwhelming?)
- Do spouses **come back** to do more? (Is it motivating and useful enough to continue?)
- Do mentors find the homework artifacts helpful in sessions?

2. How we'll test

- **Mini-surveys after each activity (30 seconds):**
- How helpful was this? Rate 1–5 stars
- Was it too long, too short, or just right?
- Would you do something like this again?
- Optional: one short comment box
- **Check simple patterns in the app:**
- Did they click "Start"?
- Did they submit something?
- Did they come back the next week?
- **Mentor feedback:**
- Quick "thumbs up / thumbs down" on whether the spouse's homework was useful.

3. How we’ll collect feedback

- **Built-in quick surveys** (after activity, anonymous by default).
- **Short weekly “confidence check-in”** (1–5 scale).
- **Mentor notes** (fast checkboxes + comment).

4. Example activities to test

- Write down 3 strengths or skills from the past week.
- Draft one STAR story about a challenge turned into a success.
- Send a networking message (template provided).

UI Link:

https:2Fsrc%2FApp.tsxo/p/sandbox/pz3jkl?
file=%2Fsrc%2FApp.tsx%3A23%2C7.

Homework Hub

Personalized activities + quick feedback to boost job-readiness.

Stage: Confidence → Career

Streak: 3 days

Assignments

Activity Player

Feedback

Strengths Journal

Confidence • 8–10 min

Not started

Write 3 moments this week that energized you. Tag skills you used.

Due in 3 days

Start

Preview

STAR Story: Relocation Win

Identity → Career • 12–15 min

In progress

Draft a STAR story that turns a challenge into a strength.

Due in 5 days

Resume

Preview

Coach

Today's nudge: Try Strengths Journal. Reflect on 3 energizing moments — I'll auto-tag strengths for you.

Micro-Network Task

Career Exploration • 5–7 min

Not started

Write 1 informational-interview request using our template.

Due tomorrow

Start

Preview

Resume Transferables

Job-Search Readiness • 10–12 min

Not started

Reframe 3 experiences into resume bullets with impact.

Due next week

Start

Preview

Quick Resources

STAR method one-pager

Networking message template

Resume impact verb list

Quick Feedback

Help us improve these activities. This takes 30–45 seconds.

How helpful was this activity?

12345

Effort required felt...

Too lowJust rightToo high

Would you recommend this to a peer?

NoMaybeYes

Anything we should change?

Optional: 1-2 lines

Anonymous by default • Toggle identity in Settings

Submit Feedback

After you submit

AI draft of resume bullet (edit anytime)

Mentor notes appear within 48h

Next activity unlocked

Thanks! Your input helps us test engagement (activation, completion, retention).

Made with Goodnotes

9/30/2025

Working with PACE in class.

The following are the available Ollama models we can work with (note the ones requiring more compute power would run slower):

```
for m in ollama.list().models:  
    print(m.model)
```

```
granite3.3:8b  
granite-code:8b  
llama3.3:70b  
llama3.2:3b  
llama3.2:1b  
llama3.1:8b  
phi4:14b  
smollm2:1.7b  
smollm2:360m  
smollm2:135m
```

Suggestions to our prototype:

- try to be more specific on the input (how to provide/format meeting summary notes to the LLM)
- note: don't solely rely on keywords, maybe consider sentiment analysis?
- we are looking at the outputs returning resources for the mentee, and homework assignments personalized to them from the meeting session
- are we leaving the resource generation to the LLM? Or is there human input in resource selection/review?

TO DO:

- attempt the process of input/output to the LLM and compare it to our intended outputs

10/14/2025

Input/Outputs:

Our system uses AI to support both mentors and mentees by automatically generating summaries, tasks, and match recommendations based on natural language inputs.

Feature 1 – Resources Based on Notes:

After each mentorship meeting, the mentor enters short notes about what was discussed. These notes are sent to the backend, which uses a LLM through an API call. The LLM reads the notes, summarizes the meeting, and generates 3–4 short, weekly action items for the spouse. These tasks are saved in JSON format. This helps mentors save time writing follow-ups and gives spouses clear next steps between sessions.

Feature 2 – Mentee–Mentor Matching:

When a new mentee creates a profile, the system uses the LLM to analyze their personality, goals, and preferred communication style. It then compares the mentee's data against a list of mentors, scoring compatibility based on shared values, personality alignment, and availability.

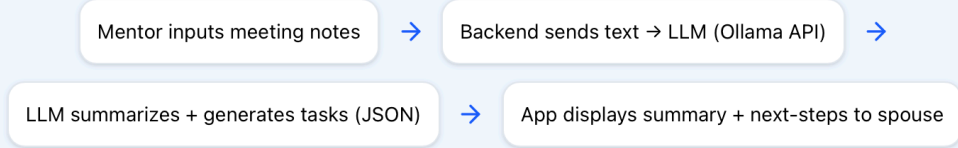
The output is a structured JSON that highlights the best match and other potential matches. This process ensures that each pairing is well-aligned with the spouse's needs.

Goal for next week:

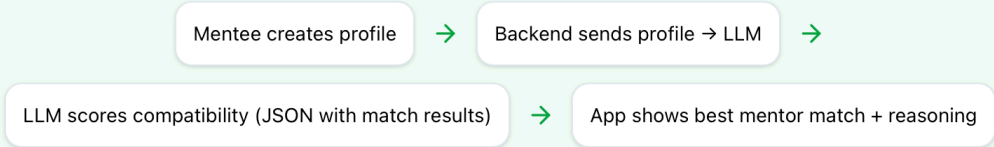
Connect with the other team.

- incorporate their chatbot into our react app?
- Move Homework Hub from sandbox to a repo

Feature 1: Resources Based on Notes



Feature 2: Mentee–Mentor Matching



10/21/2025

In class: presented updates.

Work with other subgroup on React UI to connect our components

Notes to self:

- figure out how to incorporate Ollama to AI hint feature on homework hub. Ollama analyzes military spouse resumes/ progress reports from mentor notes, and spit out possible responses to the HW assignment questions

Input to Ollama:

A json format of a military spouse's resume:

- mentor notes are optional (can be supplied to help generate hint)

```
resume_json = {
  "name": "Sarah Thompson",
  "summary": "Friendly, reflective, determined military spouse seeking flexible remote role with purpose.",
  "skills": ["Scheduling", "EMR Basics", "Customer Service", "Interview Prep", "Google Workspace", "Community Outreach"],
  "experience": [
    {
      "role": "Volunteer Coordinator",
      "organization": "Community Health Network",
      "bullets": [
        "Coordinated 20 volunteers across 3 sites; reduced scheduling conflicts by 25%.",
        "Piloted call scripts improving patient follow-up rate by 18%."
      ]
    },
    {
      "role": "Remote Tutor (Part-time)",
      "organization": "Local Education Nonprofit",
      "bullets": [
        "Delivered weekly virtual sessions; raised average learner quiz scores by 15%.",
        "Built trust with families through regular check-ins and progress notes."
      ]
    }
  ],
  "education": [
    {"degree": "B.A. Communication", "school": "State University"}
  ],
  "certifications": ["CPR", "Basic Life Support (BLS)"]
}
```

Coming up with a prompt for Ollama:

"You are an AI mentor assistant.

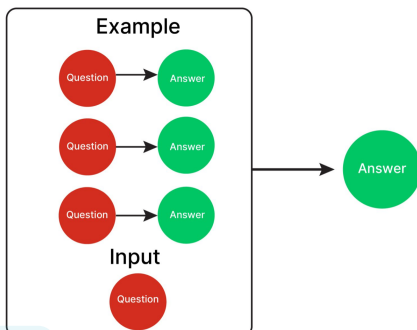
Given a homework question, a military spouse's resume (JSON), and optional mentor notes, generate 3-5 short, specific hints (under 20 words each) that help the spouse answer the homework effectively. Do not write the full answer. Focus on their actual skills/experiences, transferable strengths, and confidence-building. "

"Return STRICT JSON with keys 'hints' (array of strings) and 'rationale' (string). "

"DO NOT include markdown, code fences, labels, or any extra prose—JSON ONLY."

In addition to the prompt, I've also decided to supply it a few-shot JSON example so the LLM has a better idea of what output I want:

```
FEW_SHOT_JSON = """{
  "hints": [
    "Open with a recent example that shows impact.",
    "Name the skill used (e.g., coordination, empathy).",
    "Add one number to quantify results.",
    "Link relocation to adaptability and reliability."
  ],
  "rationale": "Concise, resume-grounded nudges that keep
responses short and confident."
}"""
```



Idea:

using few-shot prompting we can guide an AI model by including a few examples of the desired input/output within a single prompt

General Flow:

AI takes in homework question, military spouse's resume, and optional mentor notes as input. Runs getHint function using Ollama, and spits out hint of the following JSON format:

```
question = "Write three moments this week that energized you and tag skills used."
mentor_notes = "Exploring remote healthcare roles; nervous about interviews; resume refresh needed."
```

```
{
  "hints": [
    "Reflect on volunteer coordination; highlight scheduling skill.",
    "Mention tutoring success; emphasize customer service and communication.",
    "Recall a community outreach moment; focus on empathy and engagement.",
    "Quantify improvements (e.g., reduced conflicts, raised scores).",
    "Connect experiences to adaptability in remote roles."
  ],
  "rationale": "Hints leverage Sarah's skills and achievements for confidence and relevance in her resume refresh and interview preparation.",
  "used_model": "phi4:14b",
  "timestamp": "2025-10-27T22:14:57.882410+00:00"
}
```

```
question = "Draft a STAR story that turns a challenge into a strength."
mentor_notes = "Exploring remote healthcare roles; nervous about interviews; resume refresh needed."
```

```
{
  "hints": [
    "Start with your role as Volunteer Coordinator.",
    "Highlight coordination skills to address scheduling conflicts.",
    "Mention the 25% reduction in conflicts for impact.",
    "Connect adaptability from military life to remote work success."
  ],
  "rationale": "Hints leverage Sarah's experience and skills, focusing on quantifiable achievements and transferable strengths like adaptability.",
  "used_model": "phi4:14b",
  "timestamp": "2025-10-28T02:28:38.203793+00:00"
}
```

Right now, the AI is doing pretty well giving relevant hints to the user based on their resume background information.

Next step:

- expose an API in the Jupyter notebook and call it in my react app
- this way my getHint button can actually use this function implemented

11/1/2025

I've attempted to use FastAPI inside of the Jupyter notebook in order to connect the backend to my frontend react app , but keep encountering this `asyncio.run()` error:

```
File /usr/local/pace-apps/spack/packages/linux-rhel9-x86_64_v3/gcc-11.3.1/python-3.12.5-5sase6atfv2x5tf7dy5x5sqfzyguhsia/lib/python3.12/asyncio/runners.py:190, in run(main, debug, loop_factory)
    161 """Execute the coroutine and return the result.
    162
    163 This function runs the passed coroutine, taking care of
    (...) 186     asyncio.run(main())
    187 """
    188 if events._get_running_loop() is not None:
    189     # fail fast with short traceback
-> 190     raise RuntimeError(
    191         "asyncio.run() cannot be called from a running event loop")
    193 with Runner(debug=debug, loop_factory=loop_factory) as runner:
    194     return runner.run(main)

RuntimeError: asyncio.run() cannot be called from a running event loop
```

I will need to think of another way to expose an API endpoint from PACE.

Another thing I'm working on right now is making more user personas for testing my hint function. I'll combine all the resume JSON's into a separate file.

```
▼ root [] 5 items
  ▼ 0
    name "Sarah Thompson"
    summary "Friendly, reflective, determined military spouse seeking flexible remote role with purpose."
    ► skills [] 6 items
    ► experience [] 2 items
    ► education [] 1 item
    ► certifications [] 2 items
  ▼ 1
    name "Michael Reyes"
    summary "Organized, calm under pressure; seeks remote operations/logistics role supporting distributed teams."
    ► skills [] 6 items
    ► experience [] 2 items
    ► education [] 1 item
    ► certifications [] 1 item
  ▼ 2
    name "Aisha Khan"
    summary "Compassionate healthcare worker transitioning to remote/telehealth support or patient navigation."
    ► skills [] 6 items
    ► experience [] 2 items
    ► education [] 1 item
    ► certifications [] 3 items
  ► 3
  ► 4
```


Alternative to not being able to connect backend to react app yet — created a mini UI dropdown in PACE to simulate get AI hint function:

Persona: Daniel Park ▼

Question: Reframe 3 experiences into resume bullets with impact. ▼

Notes: (optional) Add mentor notes or context here...

Generate Hints

Persona: Daniel Park

Question: Reframe 3 experiences into resume bullets with impact.

HINTS:

- Highlight IT Support Specialist role: 'Closed over 50 tickets weekly, boosting first-contact resolution by 22%.'
- Focus on Volunteer Tech Lead experience: 'Implemented Wi-Fi and device access policies for over 120 users.'
- Emphasize skills used: coordination, problem-solving.
- Quantify impact with numbers to show effectiveness.
- Connect experiences to cybersecurity interest: emphasize security awareness.

Rationale: Hints guide the spouse to leverage specific achievements and skills from past roles, emphasizing quantifiable impacts and transferable strengths relevant to cybersecurity.

Timestamp: 2025-11-04T16:45:59.093904+00:00

11/10/2025

I decided to try out fastAPI locally paired with the Ollama API. Although the response call takes very long (around 30 seconds) for it to spit out a hint, but at the very least I was able to connect to our front end react app.

I also did testing on the new personas (Emily & Marcus) that my team came up with. I included them in the resumes.json file:

```
{
  "name": "Emily Chen",
  "summary": "Asian American military spouse transitioning into software development; frequent relocations",
  "skills": [
    "HTML",
    "CSS",
    "JavaScript",
    "React (beginner)",
    "Python basics",
    "Git",
    "Figma",
    "Customer Support",
    "Communication"
  ],
  "experience": [
    {
      "role": "Freelance Web Developer (Projects)",
      "organization": "Self-Directed",
      "bullets": [
        "Built responsive portfolio site with React; improved Lighthouse performance from 62 to 95.",
        "Redesigned local nonprofit homepage; raised average session duration by 28%."
      ]
    },
    {
      "role": "Customer Support Representative",
      "organization": "SaaS Helpdesk (Remote)",
      "bullets": [
        "Resolved 48+ tickets/week; 94% CSAT; wrote internal FAQ articles.",
        "Collaborated with engineering to reproduce bugs; reduced reopen rate by 18%."
      ]
    }
  ],
  "education": [
    {
      "degree": "B.S. Communication", "school": "State University"
    },
    {
      "degree": "CS/Web Dev Coursework (in progress)", "school": "Online (Coursera/edX)"
    }
  ],
  "certifications": ["freeCodeCamp Responsive Web Design (in progress)"]
}
```

```
{
  "name": "Marcus Green",
  "summary": "Senior Software Engineer passionate about mentoring nontraditional entrants; focuses on interview prep.",
  "skills": [
    "Java",
    "Python",
    "REST APIs",
    "SQL",
    "System Design",
    "Code Review",
    "Mentorship",
    "Interview Prep"
  ],
  "experience": [
    {
      "role": "Senior Software Engineer",
      "organization": "Mid-Sized Tech Company",
      "bullets": [
        "Led backend for billing service; cut API p95 latency from 480ms to 160ms.",
        "Mentored 6 junior devs; 3 promoted within 12 months."
      ]
    },
    {
      "role": "Volunteer Mentor",
      "organization": "Tech Mentorship Programs (various)",
      "bullets": [
        "Guided 18+ career switchers through projects and whiteboard practice.",
        "Ran monthly mock interviews; standardized feedback rubric to improve signal."
      ]
    }
  ],
  "education": [
    {
      "degree": "B.S. Computer Science", "school": "State University"
    }
  ],
  "certifications": ["AWS Certified Developer Associate"]
}
```

Attached is the homework hint output for Marcus using Ollama's API (phi4:14b):

AI Hints

- Use your mentoring experience to highlight a challenge.
- Describe the initial problem faced by mentees.
- Explain how you applied mentorship skills to overcome it.
- Quantify success with number of mentees or projects completed.
- Conclude by linking this experience to adaptability and leadership.

11/14/2025

New improvement:

Our teammate Geeta integrated her and Jade's mentor notes summary feature onto our frontend app.

- We also decided to switch from the Ollama API to the OpenAI API (model = gpt-4o-mini) because its response time is faster (~10 seconds) compared to the previous 30 seconds.
- Did testing on the response to check that it correctly reflects the selected personas resume information

So now our hub successfully has the get AI hint and summarize mentor notes feature integrated.

Next step: work with Anita and Tolani to get their mentor matching onto our server as well. So far it works on their end, but we want to get it integrated as a whole.

12/1/2025

New: Successfully integrated the mentor matching functionality onto the frontend react app as well. The logic was added onto the backend server.py file. As well as updated the frontend app to include the new tab for mentor matching.

Todo: look over presentation slides and add on any missing information.